MILITARY SPECIFICATION

CORE, TOROID

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contain requirements not normally covered by drawings and provide quality assurance provision for the core, toroid for use on Mine, AT, in the FASCAM Systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 Ammunition, Standard Quality Assurance Provision, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Copies of specifications, standards, drawings, publication and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.1.2 other Government documents, drawings, and Duplications. The following other Government documents, drawings, and publication form a part-of this specification to the extent-specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9287094 Core, Toroid

PUBLICATIONS

49 CFR 100-199 - Transportation, Parts

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Material shall be accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9287094

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and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Core requirements.

3.3.1 Core sensitivity. With an input drive signal of 1.0 oersted zero to peak, 10 Hertz sine waver at Z=0 the core shall produce a minimum signal of 800 millivolt, O to peak.

3.3.2 Core linearity. Core sensitivity at Z= ± 0.45 oersteds shall not vary in excess of ± 11% from the core sensitivity at Z=0 oersteds.

3.4 First article inspection. When specified in the contract or purchase order, a sample shall be subject to first article inspection in accordance with the technical provisions herein. (See 4.3)

3.5 Workmanship. The toroid core shall be fabricated, finished and tested in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defect, dirt, grease, rust, corrosion and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the part be contaminated by the cleaning agents. All markings and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVIS IONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part-of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

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4.3 First article inspection.

4. 3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of one hundred (100) toroid cores.

4.3.2 Inspection to be performed. The first article parts submitted in accordance with paragraph 4.3.1, will be subjected to all the requirements of the applicable drawings, and to the inspections and tests of Table I. See also MIL-A-48078.

4.3.3 Rejection. See MIL-A-48078. If any items fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3.4 Requalification. The device shall be subjected to another First Article as stipulated in the contract if:

a. The manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the device are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The relocation of his assembly facilities.

c. The uses of a new sub-contractor for material, parts, or assembly.

d. A change occurs in devices material, performance or assembly.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48665(AR)

PARAGRAPH	TITLE: Core, Toroid		SHEET 1 OF 1		DRAWING NUMBER: 9287094
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Core Construction	50			
	Internal Construction (Note 2D of drawing)			3.2	Visual and Microscope
	Weld strength (Note 2E of drawing)			3.2	Gage
	Marking (Note 2F of drawing)			3.2	Visual and Microscope
	Evidence of poor workmanship			3.4	Visual
	Electrical Parameter	50			
	Sensitivity at Z=0 oersteds			3.3.1	4.5.1
	Linearity at Z= -.450 oersteds			3.3.2	4.5.2
	Linearity at Z= +.450 oesteds			3.3.2	4.5.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48665(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Core, Toroid	1 OF 1	9287094
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Internal Construction (Note 2D of drawing)	32	1/	3.2	Visual and Microscope
102	Weld strength (Note 2E of drawing)	32	1/	3.2	Gage
103	Marking (Note 2F of drawing)	32	1/	3.2	Visual and Microscope
104	Sensitivity at Z=0 oersteds		100%	3.3.1	4.5.1
105	Linearity at Z= -.450 oersteds		100%	3.3.2	4.5.2
106	Linearity at Z= +.450 oersteds		100%	3.3.2	4.5.2
Minor					
201	Poor workmanship		0.65%	3.4	Visual

NOTES:

1/ Reject lot if two (2) or more defective founds.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testin Testing is decribed in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the Appendix and in the "Paragraph Reference/Inspection Method Column " of the First Article and Quality Conformance forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection. Only approved test equipment and test fixtures shall be connected to the core under test to verify that it complies with the requirements specified herein.

4.5.1 Core sensitivity . Core sensitivity shall be tested in accordance with the test fixture described in the Appendix and shall meet the requirements of 3.3.1.

4.5.2 Core linearity. Core linearity shall be tested in accordance with the test fixture described in the Appendix and shall meet the requirements of 3.3.2.

5. PACKAGING.

5.1 Preservation and packaging. The toroid core shall be preserved and packaged in accordance with MIL-STD-1169, Level C.

5.2 Packing. packing of the assemblies shall be in accordance with MIL-STD-1169. Level C.

5.3 Marking. Marking shall be in accordance with Drawing 9287094.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The core, toroid covered by this specification are intended for single-use, non-repairable military dev ice.

6.2 Ordering data. See MIL-A-48078. .

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs as required, to Commander, ARDEC, ATTN: AMSMC-QAT-I, Dover, New Jersey 07801-5001.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC), may also include drawing prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal (ARRADCOM) or Picatinny Arsenal drawings. Technical data orginally prepared by these activities is now under the cognizance of ARDEC.

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(Project 1345-A353)

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APPENDIX

10. SCOPE

10.1 Scope. This Appendix provides general information to construct a unctional test fixture to measure the core sensitivity and linearity. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS This section is not applicable to this appendix.

30. DESCRIPTION AND CONFIGURATION

30.1 General. The information defined herein includes the tests required verify the capability of the toroid core. The test tolerance on absolute magnetic values is ± 5%.

30.2 Test circuit. Figure 1 shows a block diagram of a test circuit used for core testing. This circuit consists of three sections: excitation, envelope detection, and filter and gain. It is an improvement over common signal detection circuits utilizing a flux gate magnetometer. The circuit allows the core, when subjected to the magnetic environment described in 30.4, to be tested under operating conditions that simulate actual usage.

Proper calibration will allow the test circuit to operate correctly without using standard cores for calibration. Additional circuitry however, is required to process the output from the test circuit for pass/fail indications.

30.2.1 Excitation circuit . The excitation circuit schematic, shown in Figure 2, consists of a pulse generator, drive transistors and current limiting resistors. The Darlington configuration of transistors Q1 and Q2 amplify the waveform from the pulse generator to the high current excitation signal required to saturate the 14 turn magnetometer (core under test). This current waveform is critical and must be monitored for calibration.

The current waveform can be monitored by measuring the voltage across the R2 resistor (TP2 to TPl) with TPl grounded. The current waveform of Figure 3 should be observed. The repetition rate and current amplitude are defined as:

t (pulse) = high level time = 30 us ± 10%

t (rep) = repetition time = 1 msec ± 10%

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t (rise) = less than 0.2 us

t (fall) = less than 0.2 us

I (ampl) = Current amplitude 133mA ± 5%

30.2.2 Linear envelope detector. The linear envelope detector shown in Figure 4 consists of diode D1, capacitor, C1, and resistor, R3. When the excitation signal pulse goes low (to zero) the excitation circuit appears as a high impedance with respect to the coil. The collapsing field of the coil is a negative pulse, as shown in Figure 5, which forward biases D1. C1 charges to the negative peak of each cycle of the excitation signal. When the diode is not forward biased the time constant formed by R3 and C1 allows C1 to discharge through R3 so that the output at TP3 follows the test signal envelope (10 Hz input drive signal) of the modulated (1 KHz) wave.

30.2.3 Filter and gain. The output of the linear envelope detector is first buffered by a voltage follower to ensure correct circuit separation. The signal is then input to the filter and gain section to remove the 1KHz excitation field and to amplify the low level 10 Hz output voltage of the magnetometer (core under test). Finally, the overall signal gain of 250 is set by the amplifier stage after the filter.

The filter used must meet the following parameters:

Type: Butterworth Bandpass

Order: Six

Ga in: 250

Center Frequency: Ten Hertz (tuned to the drive signal)

Q: Five

Figure 6 shows one implementation of the desired filter; it is a biquad design.

30.3 Calibration

30.3.1 Test circuit frequency calibration. The biquad filter of Figure 6 is tuned by Inputting a 40 millivolt peak to peak 10 Hz ac test signal across TP3 and ground. The output voltage at TP4 is monitored. The frequency of the signal at TP4 will be the same as that of the input signal. An adjustment is made at the first stage of the filter (R7) such that a maximum voltage is attained at TP4 (the exact amplitude being of no importance). Subsequent adjustments made at R(14), and R(21) will properly tune the second and third stages of the filter to the 10 Hz input signal.

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30.3.2 Test circuit gain calibration. Gain calibration of the filter can be accomplished by use of the circuit show in Figure 7 (Note: To use this circuit, the excitation circuit and 14 turn magnetometer must be disconnected from the test circuit at TP1) . The test signal of the circuit in Figure 7 closely approximates the flyback voltage and input signal normally seen at TP1. The test signal, as measured at TP1, shall be a 1 kilohertz, 4 volt, negative ramp with 10 hertz, 20 millivolt peak to peak modulation. For the circuit in Figure 6, the normal gain of the biquad filter is 250. R27 is adjusted so that the overall gain at TP4 is 250. That is, the output at TP4 is a 10 Hz ac signal whose peak value is 250 times the peak value of the 10 Hz ac input signal at TP3.

30.4 Magnetic environment. The magnetic environment for the core test must be resistant to externally time varying and random magnetic signals. The probe fixture is to be located at a depth equivalent to one diameter of a suitable magnetic shield. A typical implementation is shown in Figure 8. The DC bias test fields, and the 10 Hz ac drive signal field must be generated by coils within the magnetically shielded volume. All magnetic calibration measurements shall be made at the probe head for both the bias and drive signal calibration.

40. Methods of inspection.

40.1 Test geometry. Core orientation in the test fixture shall be as follows. The dot, which signifies the location of the weld mark, shall lie in the plane of the magnetometer coils and be placed as far away from the interconnect assembly as possible, as shown in Figure 9. This is to reduce the influence of the magnetic discontinuity of the weld mark. The plane of the probe head test coils shall be perpendicular to the axis of the bias and drive coils, but offset to allow the core to be at the coil’s axis to reduce magnetic field strength gradients.

40.2 Core sensitivity. Core sensitivity shall be measured using the test circuit described and calibrated herein. The magnetometer shall be oriented as shown in Figure 9. Output at TP4 shall meet the requirements of 3.3.1 with Z = O. The waveform across the 14 turns (TP1 of Figure 1 and Figure 2) should be as in Figure 5. The waveform shall reach zero volts before 30 microseconds to ensure core saturation.

40.2.2 Core linearity. Core linearity shall meet the requirements of 3.3.2 using the test described in 40.2 except Z = +0.45, and Z = -0.45 oersteds.

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50. NOTES

50.1 Definitions.

50.1.1 z. Vertical component of the earth's magnetic field, as used Z is-the simulated vertical component of the earth's magnetic field. Z is a fixed value that is time invariant.

50.1.2 Core sensitivity. The amount of filtered, amplified signal the core will pick up as a 14 turn magnetometer (formed when a core is placed-in the test fixture which provides the magnetic environment for the test circuit of Figure 1) from the calibrated 10 hertz sinusoidal signature present in the test fixture.

50.1.3 Core linearity. The percentage difference between the core sensitivity at O-bias (Z = O oersted) and N-bias (Z= +.450 oersted) or S-bias (Z = -.450 oersted)

50.1.4 Bias coils. The set of Helmholtz coils that provide the DC magnetic field (Z) for determining the DC magnetic offset for the 10hz magnetic signature.

50.1.5 Drive coils. A secondary set of Helmholtz coils that produces the 10 hz magnetic signature.

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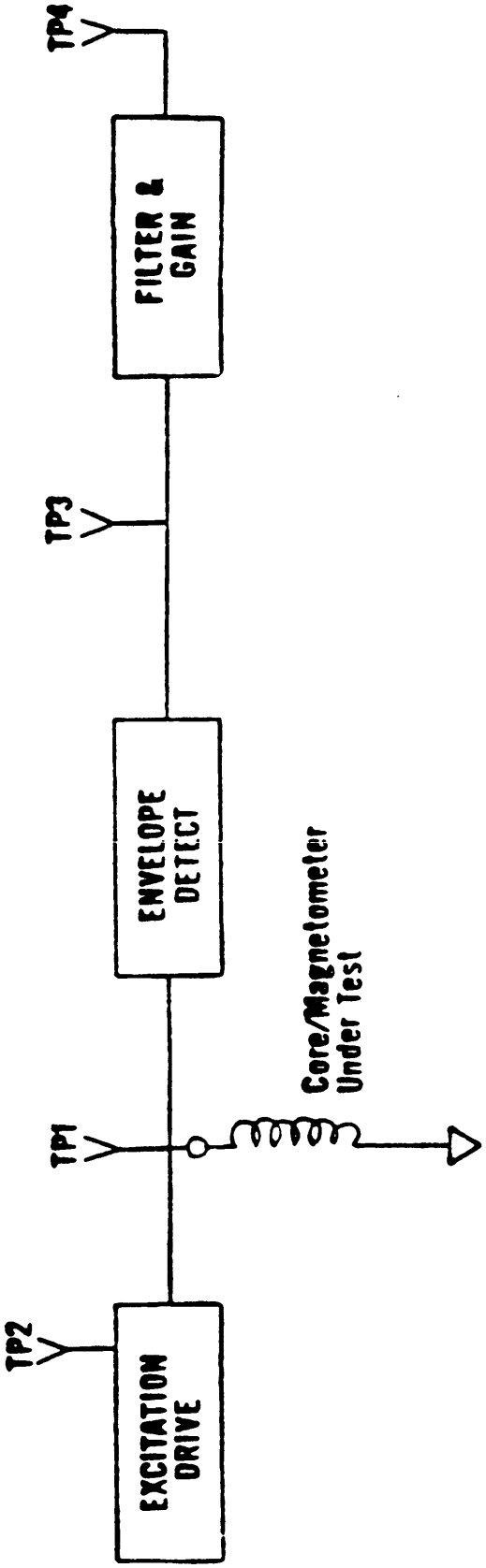


FIGURE 1. TEST CIRCUIT BLOCK DIAGRAM

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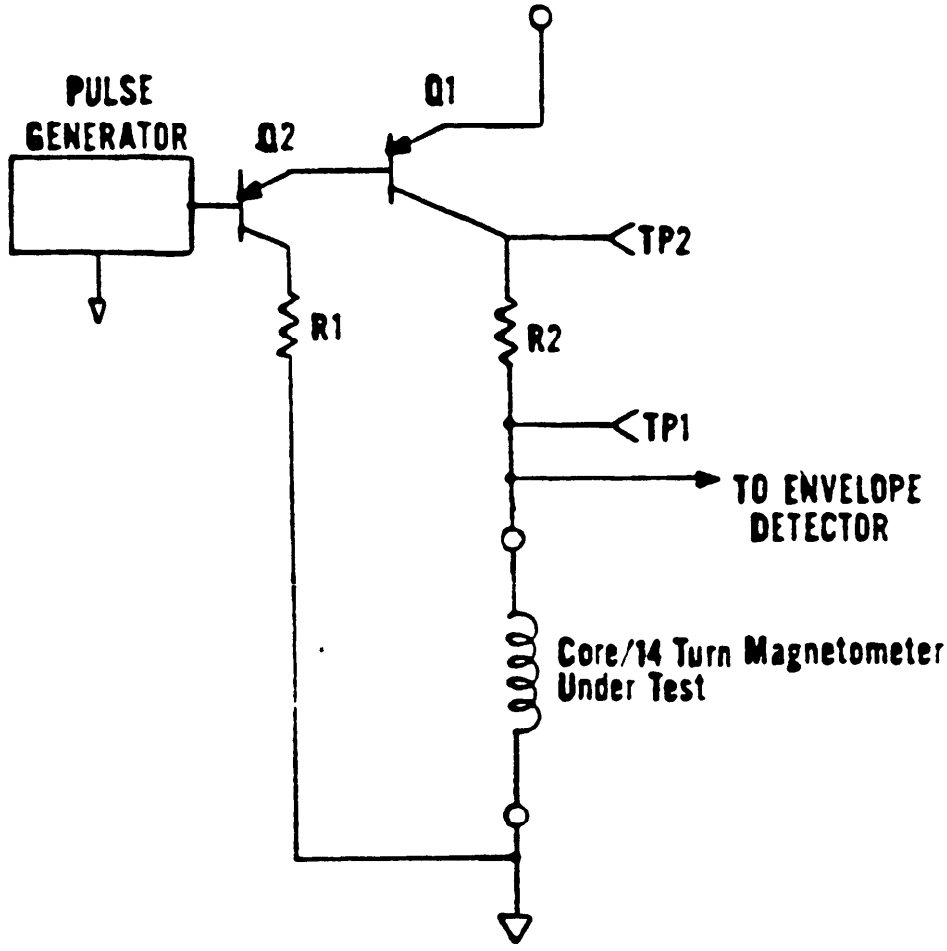


FIGURE 2. EXCITATION CIRCUIT

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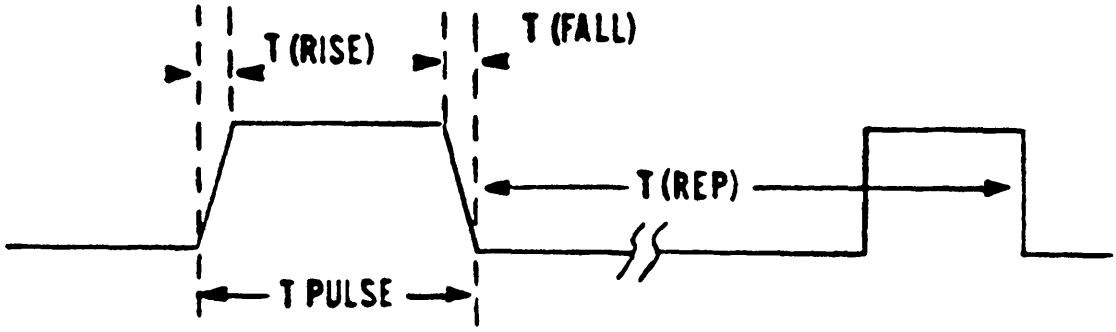


FIGURE 3. EXCITATION CURRENT WAVEFORM

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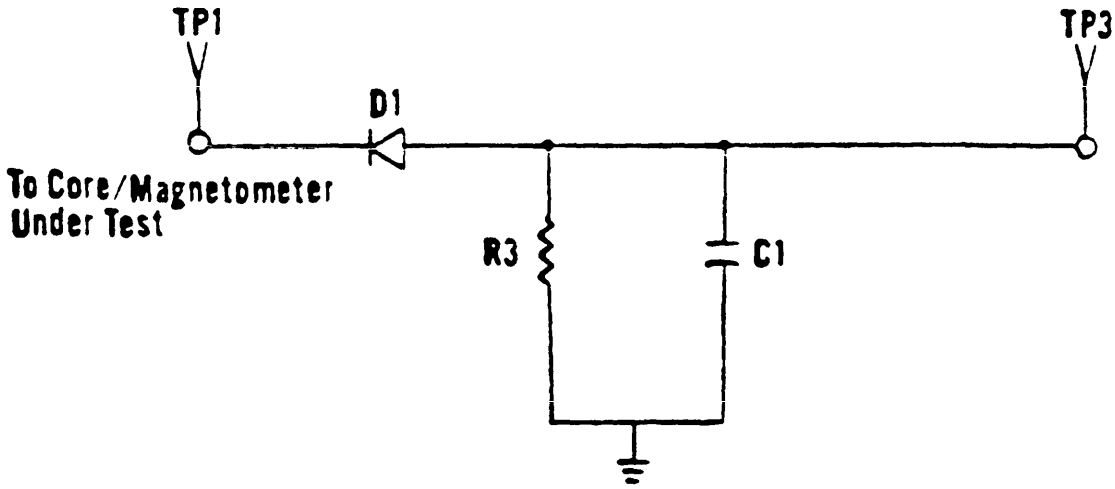


FIGURE 4. ENVELOPE DETECTOR

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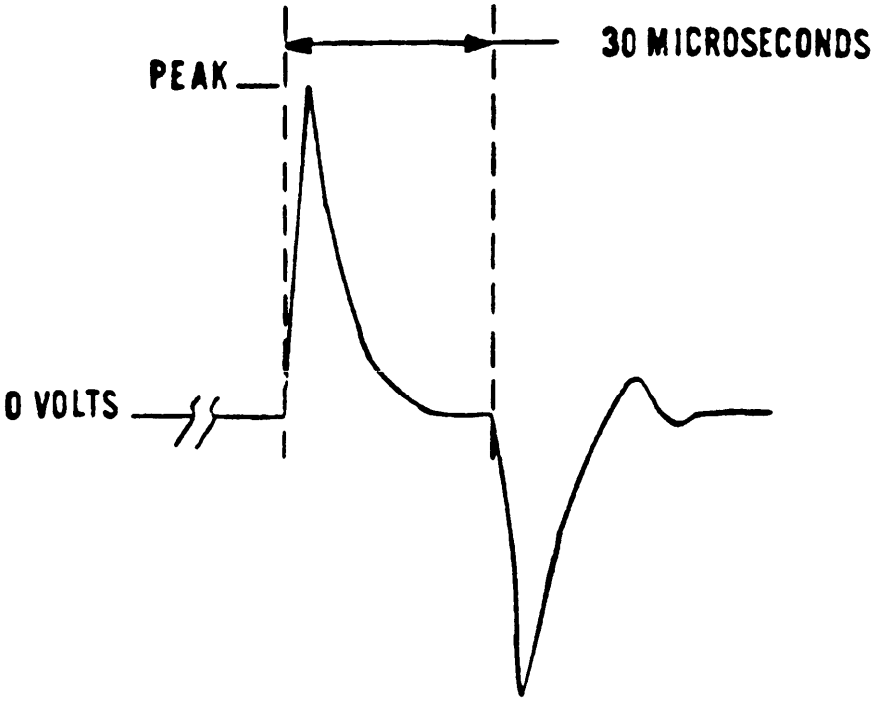


FIGURE 5. TYPICAL CORE OUTPUT AT TP1

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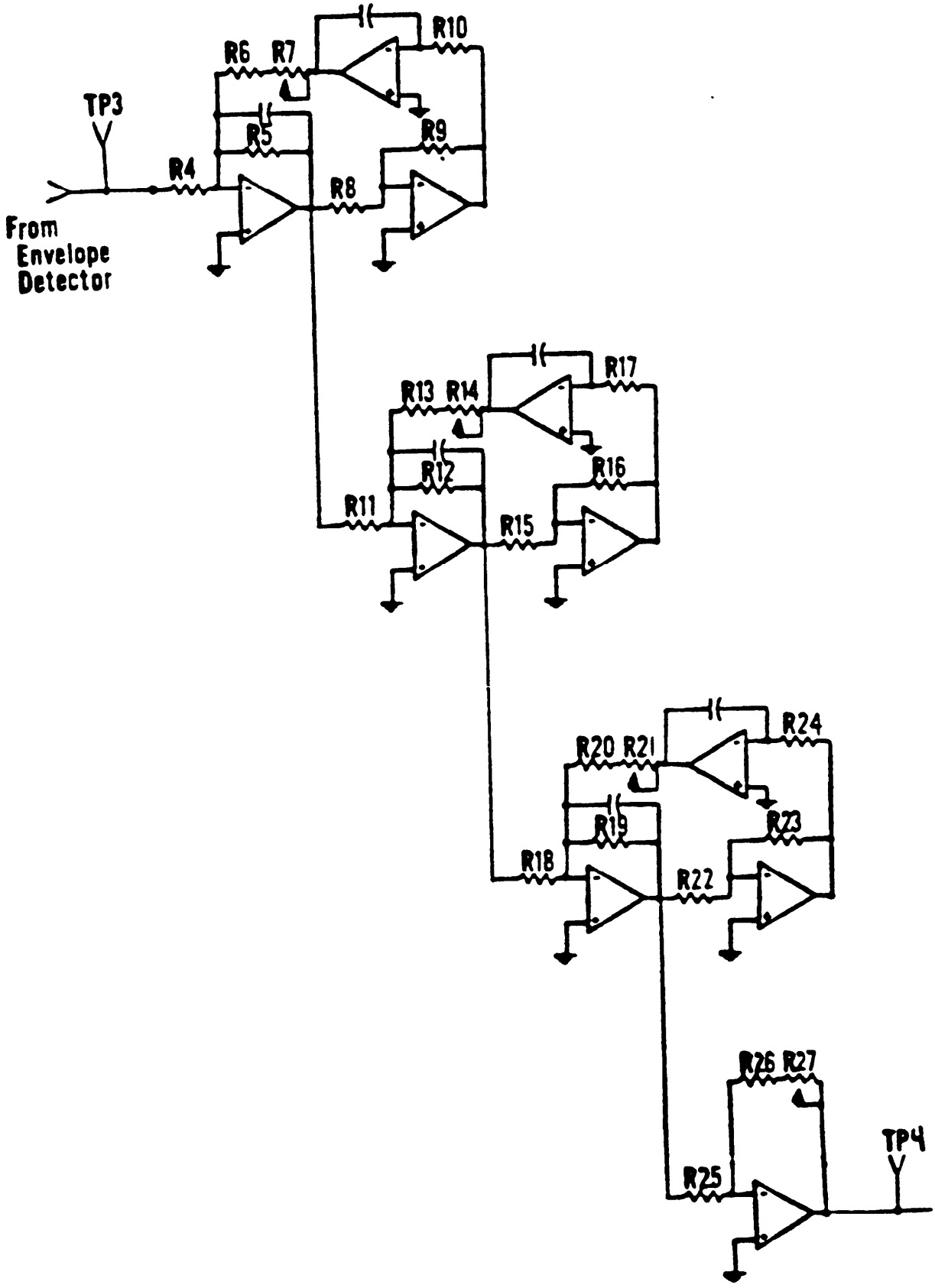


FIGURE 6. BIQUAD IMPLEMENTATION OF DESIRED FILTER

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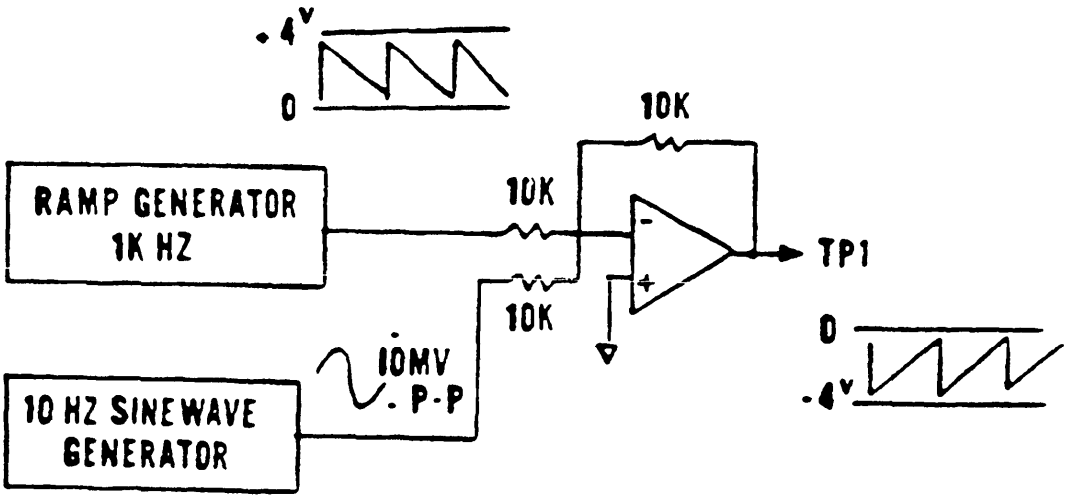


FIGURE 7 CALIBRATION TEST CIRCUIT

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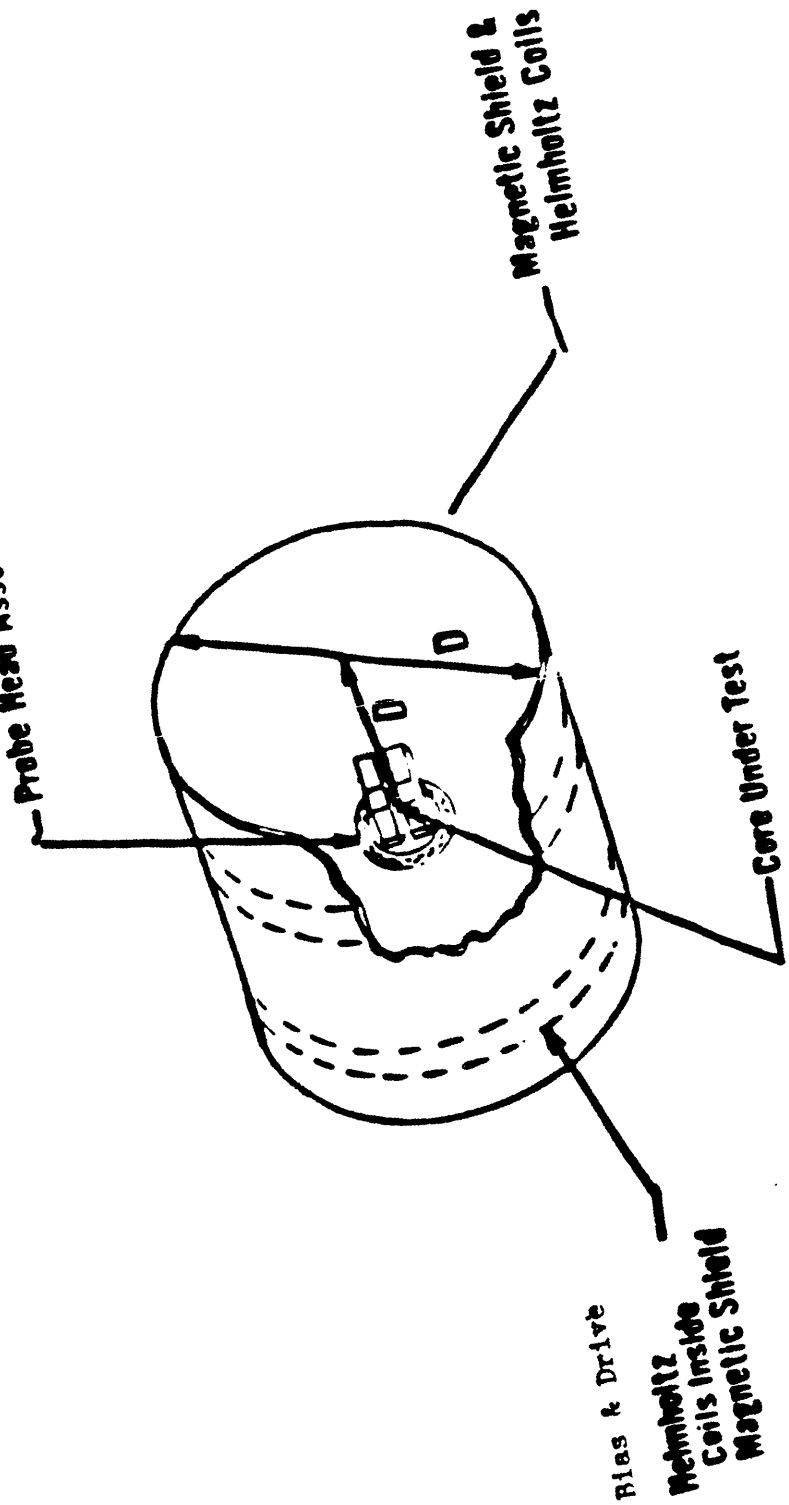


FIGURE 8
TYPICAL SHIELD AND PROBE LOCATION

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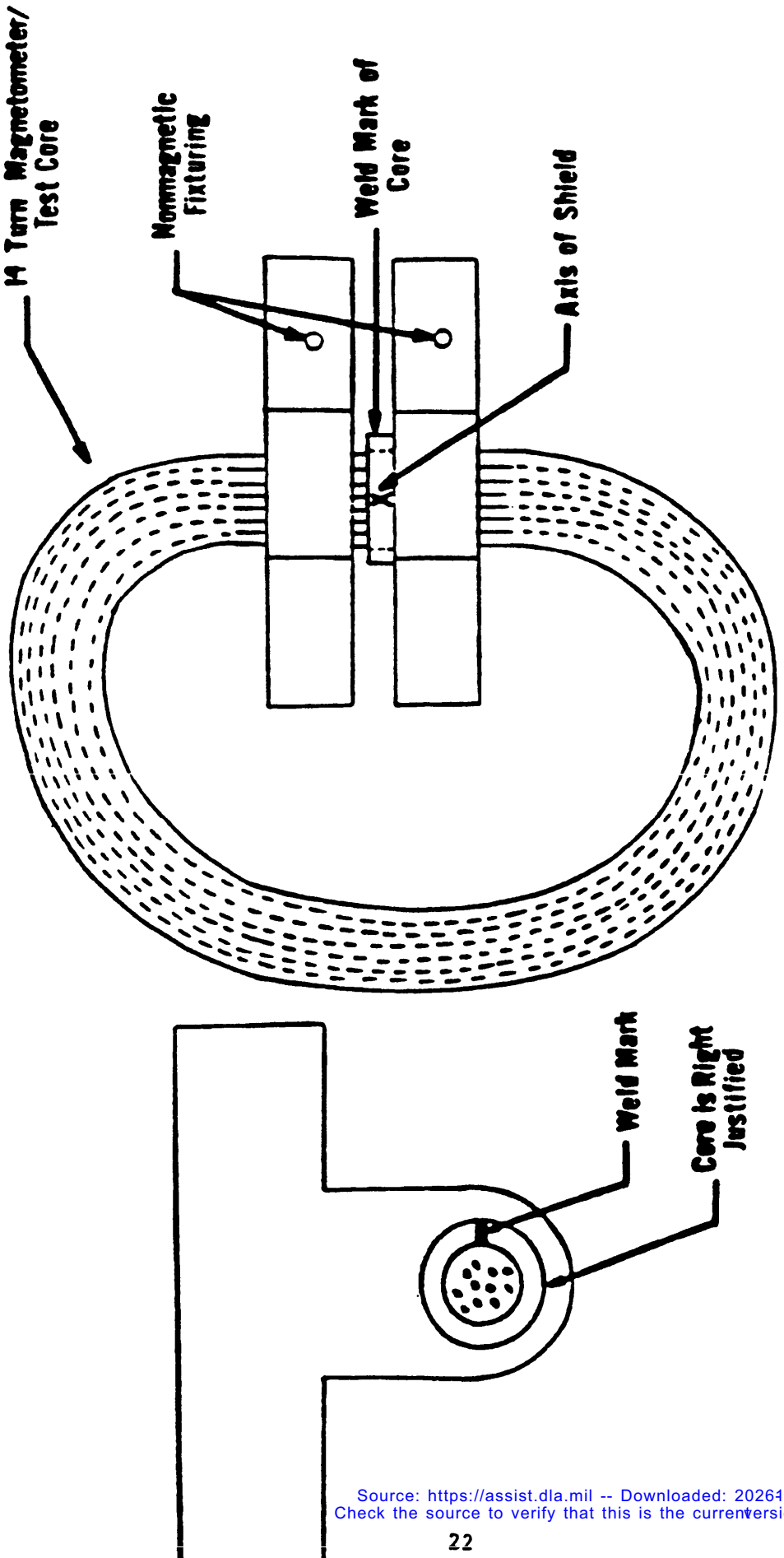


FIGURE 9. CORE POSITION WITH RELATION TO PROBE HEAD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48665	CORE, TOROID

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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